

Практикум E8

101 marks of the archive, eight techniques, twelve tasks. Everything the archive asks about the shape of a graph from May 2021 to November 2025: where the gradient is zero, what kind of point sits there, where the curve changes the way it bends, what happens when the function carries a letter, and how to show that there is no such point at all.

О практикуме

Тема взята целиком: 43 блока и 114 баллов, внутри границ гранулярности. 39 блоков и 101 балл рабочих — ноябрь 2023 года задвоился и здесь: Paper 1 TZ1 и TZ2 оказались одной бумагой (вопрос 11 совпадает слово в слово), а Paper 3 той сессии общий. Четыре блока прогнаны классификацией дважды, 13 баллов.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e8-stationary-points

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

E8 — Stationary points, concavity, inflexion

101 marks of the archive, eight techniques, twelve tasks. Everything the archive asks about the shape of a graph from May 2021 to November 2025: where the gradient is zero, what kind of point sits there, where the curve changes the way it bends, what happens when the function carries a letter, and how to show that there is no such point at all.

The one idea

A stationary point is not a formula. It is a **place on the curve**, and three questions can be asked about it, in this order:

1. **Where is it?** — solve $f'(x) = 0$, and then go back to f for the second coordinate. Two marks, and the second one is lost more often than the first.

2. **What kind is it?** — a maximum, a minimum, or a point of inflexion with zero gradient.
3. **Where does it lie?** — above or below the x -axis. This is a different question from the second, and it carries its own mark.

And what decides the kind

$$f''(a) < 0 \Rightarrow \text{maximum}, \quad f''(a) > 0 \\ \Rightarrow \text{minimum}$$

and when $f''(a) = 0$ **this test says nothing at all.** Not "inflexion" — nothing. Both $y = x^4$ (a minimum at the origin) and $y = x^3$ (an inflexion at the origin) have $f''(0) = 0$.

What always works is looking at the neighbours:

the curve next to a	the point at a
lower on both sides	a maximum
higher on both sides	a minimum
higher one side, lower the other, and $f'(a) = 0$	an inflexion with zero gradient

And a point of inflexion, stationary or not, is where the curve **stops bending one way and starts bending the other**.

How the checks work

They do not know the answers, and they do not differentiate. A curve is handed to them the way the question names it, and the check **walks along it**:

```
verify_turning('l', q1, f, 'minimum', domain=(0, 5))  
verify_nature('4b', q4b, f, [0, pi/2, pi])  
verify_bend('6a', q6a, f, domain=(-3, -0.55))
```

The first is *"is that the only local minimum of this curve with $x > 0$?"* — the check finds every place where the curve is flat and compares. The second is *"what kind of point is each of these?"* — the check looks at the neighbours. The third measures the bend with a chord.

When you are wrong the check says **how**:

what you wrote	what the check says
$(2, -64)$	the first coordinate is right and the second is not
one point where there are two	not all of them are there, the first missing one is near ...
'minimum' at a maximum	the curve is lower next to this point, not higher
-1.61 for the inflexion	there is a real point of inflexion next to that answer: check the third figure
$c \geq 0$ instead of $c > 0$	at $c = 0$ the condition says yes, and the property is not there

Order of work

level	what it means	tasks
	find the point, name its kind	1-4
	the range, the inflexion, a letter in the function	5-8
	conditions on a parameter, implicit curves, parity	9-12

Every task is a real past-paper question, cited.

70 of the 101 marks are on a calculator paper, and the calculator

earns about 13 of them. Two thirds of this topic lives in Paper 3 investigations, where the answer is a letter, a word or an inequality — and none of the three has a button.

```

import sys
sys.path.append('.')          # from
practicum/calculus to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *            # checks +
curve, stationary, nature, crossings

language('en')               # this
notebook is in English, and so are the checks

# stationary(f, domain) – every place where
the curve is flat, with its kind.
# nature(f, at) – 'maximum', 'minimum',
'inflexion' or 'neither'.
# concavity(f, at) – 'up' or 'down', measured
with a chord.
# inflexions(f, domain) – every place where
the bend changes side.
# crossings(f, domain) – how many different
points of the x-axis the curve meets.
# curve(Eq(...)) – a curve given by an
equation, walked the same way.

print('ready; sympy', sp.__version__)
sample = x**3 - 6*x**2 + 9*x + 1
print('flat places: ', stationary(sample,
(-1, 5)))
print('bend at x = 1: ', concavity(sample,
1), ' at x = 3:', concavity(sample, 3))

```

Map of the eight techniques

#	technique	you recognise it by	it reduces to
1	find the point	<i>find the coordinates of the local minimum, where the gradient is zero</i>	$f'(x) = 0$, then back to f
2	name the kind	<i>using the second derivative, show that ..., determine whether each point is a maximum or a minimum</i>	the sign of f'' at the point
3	the sign of f'	<i>state, with a reason, whether ..., by considering the sign of $f'(-1)$</i>	f' changes sign, or it does not
4	the inflexion	<i>find the x-coordinate of the point of inflexion</i>	$f'' = 0$ and the concavity changes
5	a family	<i>show that the curve has a point of zero gradient at $P(0, b)$</i>	the same work, with a letter instead of a number

#	technique	you recognise it by	it reduces to
6	which side	<i>state whether each point is above or below the x-axis</i>	the sign of the stationary value
7	how many	<i>find the set of values of c such that ...</i>	count the flat places, or the crossings
8	there are none	<i>show that the curve has no local maximum or minimum points</i>	$f'(x) = 0$ has no solution in the domain

Techniques 1–4 are one point read four ways. Technique 5 changes nothing but the alphabet. Techniques 6–8 are all the same question in disguise: **how many of these points are there, and where do they sit** — and the answer is a set, an inequality or the empty list.

Part 1. Find it, and name it

Theory: the two coordinates

Take $f(x) = x^3 - 6x^2 + 9x + 1$. Then $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, so the graph is flat at $x=1$ and at $x=3$. That is **half the answer**. A point has two coordinates, and the second one comes from f , never from f' :

$$\begin{aligned}f(1) &= 1 - 6 + 9 + 1 \\&= 5, \quad f(3) \\&= 27 - 54 + 27 + 1 \\&= 1\end{aligned}$$

so the stationary points are $(1,5)$ and $(3,1)$.

The mark that goes missing. "Find the coordinates of the local minimum point" is two marks in nearly every paper: one for x , one for y . Writing $x=3$ and stopping loses the second. Writing $f'(3)=0$ as the second coordinate loses it too — that number

is zero at every stationary point and says nothing.

Which of them is which? Between $x = 1$ and $x = 3$ the factor $(x - 1)$ is positive and $(x - 3)$ is negative, so $f' < 0$ there: the curve goes down from $(1, 5)$ to $(3, 1)$. The first is a maximum, the second a minimum.

A restricted domain adds its own flat places. On a closed interval the smallest and largest values of f can sit at the ends, where f' is not zero at all. The question will say which it wants: *stationary points* means $f' = 0$; *the range* means the ends count too.

Task 1 — **May 2025 TZ1 Paper 1 Q1, 5 marks**

Consider the function

$$f(x) = \frac{4x^3}{3} - 16x, \quad x \in \mathbb{R}$$

The graph of $y = f(x)$ has a local minimum point at (p, q) where $p > 0$.

Find the value of p and the value of q .

Enter the point as a pair. The answer is exact — no calculator was allowed.

```
q1 = ...           # the point (p, q)

f = 4*x**3/3 - 16*x

verify_turning('l', q1, f, 'minimum', domain=
(0, 6))
```

Theory: when the calculator does the finding

Two marks, one screen. `fMin`, `fMax` or the graph's own *maximum* command gives both coordinates at once, and the answer is written to three significant figures.

For $y = \frac{x}{x^2 + 4}$ on $-6 \leq x \leq 6$ the calculator returns $(2, 0.25)$ and $(-2, -0.25)$ — and here the exact values happen to be short, which is the exception. Usually they are not, and the paper asks for three figures on purpose.

Two things the screen will not tell you.

- **Which is which.** The calculator labels a maximum on the piece of graph it can see. On a curve with two humps, the *local* maximum you are asked about may not be the highest point on the screen.
- **Whether you found them all.** Zoom out before you answer. A stationary point outside the window is still in the domain.

Round only at the end. Feed the calculator's own stored value back into f for the second coordinate; typing in the rounded x can move the third significant figure of y .

Task 2 ● — November 2022 Paper 2 Q2(a) and May 2025 TZ2 Paper 3 Q1(a)(iii), 4 marks

(a) The function f is defined as $f(x) = \ln(xe^x + 1) - x^4$ for $0 \leq x \leq 2$. The graph of f has a local maximum at point A . Find the coordinates of A .

(b) Consider the curve given by

$$y = \frac{x(x^2 - 16)}{x^2 + 16}$$

State the coordinates of the local maximum point and the coordinates of the local minimum point.

Three significant figures throughout. In (b) give both points as a list.

```
q2a = ...          # the coordinates of A
q2b = [...]        # the local maximum and the
                    local minimum

first = log(x*exp(x) + 1) - x**4
second = x*(x**2 - 16)/(x**2 + 16)

verify_turning('2(a)', q2a, first, 'maximum',
domain=(0, 2))
verify_turning('2(b)', q2b, second, domain=
(-10, 10))
```

Theory: the sign of f' , read off a diagram

Sometimes the derivative is never computed at all — you are handed its sign.

x	...	-1	...	2	...
$f'(x)$		-		0	+

The gradient goes from negative to positive at $x = 2$: the curve falls, flattens and rises, so $x = 2$ carries a **local minimum**. That is the whole argument, and in the markscheme it is worth as much as the word itself — *state, with a reason* means the reason is a separate mark.

The same reading for f'' . Where f'' changes sign the curve changes the side it bends to, and that is a **point of inflexion**.

$f''(c) = 0$ **is not a reason.** A markscheme that asks for a point of inflexion awards R0 for "because $f''(c) = 0$ ", and awards the mark for "because f'' changes sign at c ". The curve $y = x^4$ is the reason why: $f''(0) = 0$ and the origin is a minimum, not an inflexion.

The three words to have ready: *changes from positive to negative* (maximum), *from negative to positive* (minimum), *changes sign* (inflexion).

Task 3 ● — **November 2025 TZ1**
Paper 1 Q9(b) and (d), 4 marks

The function f has a derivative given by $f'(x) = 3x^2 + 12x - 15$.

The graph of $y = f(x)$ has horizontal tangents at the points where $x = a$ and $x = b$, with $a < b$; part (a) gives $a = -5$ and $b = 1$. The sign of f' in the three intervals they cut is

+ a − b +

(b) State, with a reason, whether there is a local maximum point or a local minimum point on the graph of $y = f(x)$ at $x = a$.

The second derivative $f''(x)$ is zero at $x = c$; part (c) gives $c = -2$, and the sign of f'' on either side of it is − then +.

(d) State, with a reason, whether there is a point of inflexion on the graph of $y = f(x)$ at $x = c$.

Enter one word each: 'maximum', 'minimum', 'inflexion' or 'neither'. Only f' is given, so the cell rebuilds an f from

it — the shape does not depend on the constant.

```
q3b = ...           # the kind of point at x = a
q3d = ...           # the kind of point at x = c

slope = 3*x**2 + 12*x - 15
shape = integrate(slope, x)           # any
antiderivative: the shape is the same

verify_nature('3(b)', q3b, shape, -5, domain=
(-9, 5))
verify_nature('3(d)', q3d, shape, -2, domain=
(-9, 5))
```

Theory: the second derivative, and where it goes quiet

Take $f(x) = x^4 - 8x^2$. Then $f'(x) = 4x^3 - 16x = 4x(x^2 - 4)$, flat at $x = 0$ and $x = \pm 2$, and $f''(x) = 12x^2 - 16$:

$$\begin{aligned} f''(0) &= -16 < 0 \\ &\Rightarrow \text{maximum,} & f''(\pm 2) \\ &= 32 > 0 \\ &\Rightarrow \text{minimum} \end{aligned}$$

Three marks, three substitutions. **Write down the value, not just the sign** — markschemes for this technique say *the*

values for the second derivative must be correct in order to award the R marks.

And now the case the test cannot do.

For $g(x) = x^4$ we get $g'(0) = 0$ and $g''(0) = 0$. The test is silent, and the answer is still a minimum — $x^4 > 0$ on both sides of the origin. For $h(x) = x^5$, again $h'(0) = h''(0) = 0$, and this time the origin is a point of inflexion with zero gradient.

	x^4	x^5
$f''(0)$	0	0
left of 0	above	below
right of 0	above	above
the origin is	a minimum	an inflexion

The neighbours separate the two cases; the second derivative does not. That is why the checks in this notebook walk the curve.

Task 4 ● — November 2023 TZ2 Paper 1 Q11(a) and (b), 9 marks

Consider the function

$$f(x) = e^{\cos 2x}, \quad -\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$$

(a) Find the coordinates of the points on the curve $y = f(x)$ where the gradient is zero.

(b) Using the second derivative at each point found in part (a), show that the curve $y = f(x)$ has two local maximum points and one local minimum point.

Exact coordinates in (a) — this is Paper 1. In (b) give the three kinds as a list, in the same order as the points.

```
q4a = [...]      # every point where the
                  # gradient is zero
q4b = [...]      # the kind of each, in the
                  # same order

f = exp(cos(2*x))
window = (-pi/4, 5*pi/4)

verify_turning('4(a)', q4a, f, domain=window)
verify_nature('4(b)', q4b, f, [0, pi/2, pi],
              domain=window)
```

Part 2. The range, the bend, and a letter in the function

Theory: stationary points give the range

The range of f on a closed interval is decided by four numbers at most: the two ends and the stationary values between them.

For $f(x) = x\sqrt{4-x^2}$ on $-2 \leq x \leq 2$ — a different function from the one below, and a shorter one —

$$f'(x) = \sqrt{4-x^2} - \frac{x^2}{\sqrt{4-x^2}} = \frac{4-2x^2}{\sqrt{4-x^2}}$$

which is zero when $x = \pm\sqrt{2}$, and $f(\pm\sqrt{2}) = \pm 2$. At the ends $f(\pm 2) = 0$. So the range is $-2 \leq y \leq 2$.

Three separate marks live here: the differentiation (product and chain rules together), the solving, and the substitution back into f . Stopping at $x = \pm\sqrt{2}$ answers a question nobody asked.

A quotient that is zero. $\frac{4 - 2x^2}{\sqrt{4 - x^2}} = 0$

needs only the numerator to vanish — but the denominator must still be alive there. At $x = \pm 2$ the fraction is undefined, not zero, and those are ends of the domain, not stationary points.

Task 5 — **May 2022 TZ2 Paper 1 Q6(b), 6 marks**

A function f is defined by

$$f(x) = x\sqrt{1 - x^2}, \quad -1 \leq x \leq 1$$

The range of f is $a \leq y \leq b$, where $a, b \in \mathbb{R}$.

Find the value of a and the value of b .

*Enter the range itself, as an interval:
Interval(a , b). Exact values.*

```
q5 = ...           # the range, as an interval

f = x*sqrt(1 - x**2)

verify_range('5', q5, f, domain=Interval(-1, 1))
```

Theory: the point of inflexion

A point of inflexion is where the curve **changes the side it bends to**. Two things must both be true, and the second is the one that carries the mark:

$$f''(x) = 0 \quad \text{and} \quad f'' \text{ changes sign there}$$

Take $f(x) = \frac{x}{x^2 + 1}$. A little work gives

$$f''(x) = \frac{2x(x^2 - 3)}{(x^2 + 1)^3}, \text{ which vanishes at } x = 0$$

and $x = \pm\sqrt{3}$ — and at each of the three the sign really does change, so all three are points of inflexion.

On a calculator paper there is a shorter route, and the markscheme names it: graph $\frac{dy}{dx}$ and find *its* maximum or minimum. Where the gradient is steepest or flattest, the curve is turning over — that is the inflexion, and one screen gives it.

$$\begin{aligned} \text{inflexion of } f &\iff \text{stationary point of } f' \\ &\iff \text{zero of } f'' \end{aligned}$$

A zero of f'' need not be an inflexion. For $f(x) = x^4$, $f''(0) = 0$ and the curve bends upward on both sides. The sign has to turn over, not merely touch zero.

Task 6 — **May 2021 TZ1 Paper 2 Q11(c) and May 2023 TZ2 Paper 2 Q12(e), 4 marks**

(a) The function f is defined by

$$f(x) = \frac{3x + 2}{4x^2 - 1}, \quad x \neq \pm \frac{1}{2}$$

The graph of $y = f(x)$ has exactly one point of inflexion. Find its x -coordinate.

(b) The graph of

$$y = x\sqrt{\frac{9x^4 - 1}{2}}, \quad \frac{\sqrt{3}}{3} < x < 1$$

has a point of inflexion at the point P . By sketching the graph of an appropriate derivative of y , determine the x -coordinate of P .

Three significant figures. The x -coordinate only — that is all the question asks for.

```

q6a = ...          # the x-coordinate of the
point of inflexion
q6b = ...          # the x-coordinate of the
inflexion of the second curve

first = (3*x + 2)/(4*x**2 - 1)
second = x*sqrt((9*x**4 - 1)/2)

verify_bend('6(a)', q6a, first, domain=(-4,
-0.55))
verify_bend('6(b)', q6b, second, domain=
(sqrt(3)/3 + 0.001, 1))

```

Theory: the same work with a letter

Nothing changes when a parameter replaces a number — except that the answer is an expression and the words *show that* appear.

Take $y = x^3 + kx$ for $k < 0$. Then $\frac{dy}{dx} = 3x^2 + k$, which is zero when $x = \pm\sqrt{-k/3}$, and substituting back,

$$\begin{aligned}
 y &= \left(\pm\sqrt{-\frac{k}{3}}\right)^3 + k\left(\pm\sqrt{-\frac{k}{3}}\right) \\
 &= \pm\left(-\frac{k}{3}\right)^{3/2} \mp k\sqrt{-\frac{k}{3}} \\
 &= \mp\frac{2k}{3}\sqrt{-\frac{k}{3}}
 \end{aligned}$$

The sign of k is the whole question.

For $k < 0$ there are two stationary points; for $k > 0$ the equation $3x^2 = -k$ has no real solution and there are none; for $k = 0$ the two merge into one, and it is an inflexion with zero gradient.

Classifying a family. $\frac{d^2y}{dx^2} = 6x$, so the point at $x = -\sqrt{-k/3}$ has $\frac{d^2y}{dx^2} < 0$ and is a maximum, and the one at $+\sqrt{-k/3}$ is a minimum. Notice what was needed: not the value of k , only its sign.

"Show that" means arrive at the printed line. The markscheme gives AG for the last step only if it is written out. A correct expression obtained but never simplified into the form asked for loses the mark.

Task 7 — November 2023 Paper 3 Q1(e), (f) and (g)(ii), 10 marks

This question explores the family of curves

$$y = x^3 + ax^2 + b, \quad x \in \mathbb{R}, a \neq 0, b \in \mathbb{R}$$

(e) Show that the curve has a point of zero gradient at $P(0, b)$ and a point of zero gradient at $Q\left(-\frac{2a}{3}, \frac{4a^3}{27} + b\right)$.

(f) Consider the points P and Q for $a > 0$ and $b > 0$.

(i) Determine whether each point is a local maximum or a local minimum.

(ii) Determine whether each point is located above or below the x -axis.

(g) Consider the points P and Q for $a < 0$ and $b > 0$. State the conditions on a and b that determine when Q is below the x -axis.

In (e) give both points as a list, in terms of a and b . In (f) give two words each, P first. In (g) write the condition as an inequality.

```

a, b = symbols('a b')

q7e = [...]      # the two points of zero
gradient, in terms of a and b
q7f_i = [...]    # 'maximum' or 'minimum' for
the two points, in order
q7f_ii = [...]   # 'above' or 'below' for the
two points, in order
q7g = ...        # the condition on a and b
for Q to be below the axis

y_curve = x**3 + a*x**2 + b
positive = {a: (3, 1, 2, 6), b: (1, 2, 5, 1)}
# a > 0, b > 0

verify_turning('7(e)', q7e, y_curve, domain=
(-8, 4), params=positive)
verify_nature('7(f)(i)', q7f_i, y_curve, [0,
-2*a/3], domain=(-8, 4), params=positive)
verify_side('7(f)(ii)', q7f_ii, y_curve, [0,
-2*a/3], params=positive)

def below(a, b):
    place = [spot for spot in stationary(x**3
+ a*x**2 + b, (-20, 20))
              if abs(spot[0]) > 1e-6]
# this is Q, the one away from 0
    return None if not place else place[0][1]
< 0

verify_condition('7(g)', q7g, below,
symbols('a b'), window=(-3, -0.25), steps=10)

```

Theory: showing a maximum when everything is a letter

Two kinds of question look alike and are not.

"Determine whether" wants the sign of f'' at the point and a word.

"Use f'' to show that there is a local maximum at $x = m$ " wants the sign established *for every allowed value of the letters*, and that usually means simplifying $f''(m)$ until its sign is visible.

Take $V = x(6 - 2x)(k - 2x)$ with $k > 3$, so that $V' = 12x^2 - 4(3 + k)x + 6k$ and $V'' = 24x - 4(3 + k)$. The smaller root of $V' = 0$ is

$$x_m = \frac{(3 + k) - \sqrt{9 - 3k + k^2}}{6}$$

and substituting it,

$$\begin{aligned} V''(x_m) &= 24 \cdot \frac{(3 + k) - \sqrt{9 - 3k + k^2}}{6} - 4(3 + k) \\ &= -4\sqrt{9 - 3k + k^2} \end{aligned}$$

A square root is never negative, so $V''(x_m) < 0$ **whatever k is**, and the point is a maximum. The $(3 + k)$ cancelling is

the whole trick: what is left carries its sign on its face.

A local maximum is not automatically the largest value. On a closed interval the ends must still be checked. Here $V = 0$ at both ends and $V > 0$ inside, so the local maximum is the global one — and that is a separate mark in the markscheme.

Task 8 — **May 2025 TZ3 Paper 3**
Q2(d), 5 marks

A rectangular sheet of cardboard measures a cm by b cm, where $a < b$. A square of side x cm is cut from each corner and the sheet is folded into an open-topped box, so that

$$V = x(a - 2x)(b - 2x), \quad 0 \leq x \leq \frac{a}{2}$$

Part (c) gives that the only solutions of $\frac{dV}{dx} = 0$ are

$$x = \frac{(a + b) \pm \sqrt{a^2 - ab + b^2}}{6}$$

and

$$x_m = \frac{(a + b) - \sqrt{a^2 - ab + b^2}}{6}$$

is the one inside $0 \leq x \leq \frac{a}{2}$.

(i) Find an expression for $\frac{d^2V}{dx^2}$.

(ii) Use it to show that there is a local maximum of V at $x = x_m$.

In (ii) enter one word.

```
a, b = symbols('a b', positive=True)

q8_i = ...      # d^2V/dx^2 in terms of a, b
and x
q8_ii = ...     # the kind of point at x_m

V = x*(a - 2*x)*(b - 2*x)
x_m = ((a + b) - sqrt(a**2 - a*b + b**2))/6
sheets = {a: (3, 2, 4, 5), b: (5, 7, 9, 6)}
# a < b, both positive

verify_derivative('8(i)', q8_i, V, order=2)
verify_nature('8(ii)', q8_ii, V, x_m, domain=
(0, a/2), params=sheets)
```

Part 3. How many, and where

Theory: counting the flat places

"For which values of the parameter does the graph have two stationary points?" is not a question about points at all. It is a question about **how many solutions** $f'(x) = 0$ **has**, and it is answered by the discriminant.

For $f(x) = x^3 + px^2 + 3x$ we get $f'(x) = 3x^2 + 2px + 3$, a quadratic, and

$$\Delta = 4p^2 - 36$$

Δ	the equation $f' = 0$	the graph
> 0 , that is \$	p	> 3 \$
$= 0$, that is $p = \pm 3$	one repeated root	one inflexion with zero gradient
< 0 , that is \$	p	< 3 \$

Read the middle row carefully. A repeated root of f' means f' **touches**

zero without crossing it, so the gradient keeps its sign on both sides: the curve flattens for an instant and goes on. That is precisely a point of inflexion with zero gradient, and it is why the three rows above are the three answers the exam wants.

Write the answer as a set. *The set of values of p is $|p| > 3$, or $p < -3$ or $p > 3$ — not " $p = 4$ works". One example is not a set.*

Task 9 — **May 2021 TZ1 Paper 3 Q1(c) and (d), 8 marks**

This question explores cubic polynomials of the form $x^3 - 3cx + d$. Consider

$$f(x) = x^3 - 3cx + 2, \quad x \in \mathbb{R}, c \in \mathbb{R}$$

Part (b) gives $f'(x) = 3x^2 - 3c$.

(c) Hence, or otherwise, find the set of values of c such that the graph of $y = f(x)$ has

(i) a point of inflexion with zero gradient;

(ii) one local maximum point and one local minimum point;

(iii) no points where the gradient is equal to zero.

(d) Given that the graph of $y = f(x)$ has one local maximum point and one local minimum point, show that the y -coordinate of the local maximum point is $2c^{3/2} + 2$ and that the y -coordinate of the local minimum point is $-2c^{3/2} + 2$.

In (c) enter sets: `FiniteSet(0)`, `Interval.open(0, oo)`, and so on. In (d) give both points, maximum first, in terms of `c`.

```

c = symbols('c')

q9c_i = ...      # the set of c with an
inflexion of zero gradient
q9c_ii = ...     # the set of c with one
maximum and one minimum
q9c_iii = ...    # the set of c with no point
of zero gradient
q9d = [...]      # the local maximum and the
local minimum, in terms of c

family = x**3 - 3*c*x + 2

def kinds(value):
    return [word for _, _, word in
stationary(family.subs(c, value), (-5, 5))]

verify_param_set('9(c)(i)', q9c_i, lambda v:
kinds(v) == ['inflexion'],
                var=c, window=(-4, 4))
verify_param_set('9(c)(ii)', q9c_ii, lambda
v: kinds(v) == ['maximum', 'minimum'],
                var=c, window=(-4, 4))
verify_param_set('9(c)(iii)', q9c_iii, lambda
v: kinds(v) == [],
                var=c, window=(-4, 4))

verify_turning('9(d)', q9d, x**3 - 3*c*x + 2,
domain=(-4, 4),
                params={c: (1, 4, Rational(1,
4), 2)})

```

Theory: from the stationary values to the number of roots

A cubic crosses the x -axis once, twice or three times, and the stationary values decide which. Let the maximum value be M and the minimum value be m , with $M > m$:

	the graph meets the axis
$m > 0$ or $M < 0$ (both vertices on one side)	once
$m = 0$ or $M = 0$ (a vertex sits on the axis)	twice
$m < 0 < M$ (the vertices straddle the axis)	three times

The middle row is the one to think about. A vertex on the axis is a **repeated root**: the curve touches there instead of crossing, and touching counts as one intersection, not two and not none.

For $y = x^3 - 3x + k$ the vertices are at $x = \pm 1$ with values $k + 2$ and $k - 2$, so the three rows read $|k| > 2$, $|k| = 2$, $|k| < 2$.

And when the vertices themselves move. If both c and d vary in $x^3 - 3cx + d$, the case $c \leq 0$ has to be handled first and separately: there are no vertices at all, the curve is increasing throughout, and one crossing is forced whatever d is. Only for $c > 0$ does the comparison above begin.

State the case, not only the answer.

Six marks of a Paper 3 went on exactly this question, and the markscheme gives at most half of them if " $c > 0$ " is never written down.

Task 10 — **May 2021 TZ1 Paper 3 Q1(e) and (f), 12 marks**

Still with $f(x) = x^3 - 3cx + 2$, and with the y -coordinates of the vertices found in task 9.

(e) Hence, for $c > 0$, find the set of values of c such that the graph of $y = f(x)$ has

- (i)** exactly one x -axis intercept;
- (ii)** exactly two x -axis intercepts;

(iii) exactly three x -axis intercepts.

Now consider $g(x) = x^3 - 3cx + d$ for $x \in \mathbb{R}$, where $c, d \in \mathbb{R}$.

(f) Find all conditions on c and d such that the graph of $y = g(x)$ has exactly one x -axis intercept.

In (f) write one condition joining the cases: $0r(\dots, \dots, \dots)$.

```

c, d = symbols('c d')

q10e_i = ...      # the set of c with exactly
one x-axis intercept
q10e_ii = ...     # the set of c with exactly
two
q10e_iii = ...    # the set of c with exactly
three
q10f = ...        # all conditions on c and d
for exactly one intercept

def meets(value, times):
    if value <= 0:
        return None                                #
part (e) speaks only of  $c > 0$ 
    return crossings(x**3 - 3*value*x + 2,
(-30, 30)) == times

verify_param_set('l0(e)(i)', q10e_i, lambda
v: meets(v, 1), var=c, window=(0, 4))
verify_param_set('l0(e)(ii)', q10e_ii, lambda
v: meets(v, 2), var=c, window=(0, 4))
verify_param_set('l0(e)(iii)', q10e_iii,
lambda v: meets(v, 3), var=c, window=(0, 4))

def once(c, d):
    return crossings(x**3 - 3*c*x + d, (-30,
30)) == 1

verify_condition('l0(f)', q10f, once,
symbols('c d'), window=(-3, 3), steps=12)

```

Theory: a curve that is not a function

When the curve is given by an equation in x and y , the gradient comes from implicit differentiation, and everything else is unchanged.

For $x^2 + xy + y^2 = 7$,

$$\begin{aligned}2x + y + x \frac{dy}{dx} + 2y \frac{dy}{dx} &= 0 \\ \implies \frac{dy}{dx} &= -\frac{2x + y}{x + 2y}\end{aligned}$$

A fraction is zero when its numerator is — and its denominator is not. So $y = -2x$, and substituting into the curve, $x^2 - 2x^2 + 4x^2 = 7$, giving $x = \pm\sqrt{7/3}$ and two stationary points.

And a fraction is never zero when its numerator cannot be. That is how *show that there are no stationary points* is done: set the numerator to zero and find that no point of the curve satisfies it — either because the equation has no

real solution, or because its solutions are outside the domain the question states.

The domain does the work. Half the marks of such a question are for noticing that the candidate solutions are excluded. " $y = \pm 2$, and the question says $-2 < y < 2$, so there are none" is the whole argument, and the last clause is the mark.

Task 11 — **May 2022 TZ2 Paper 3 Q1(d)(ii) and (e), 8 marks**

Consider the curve $y^2 = x^3 + x$ for $x \geq 0$. Part (d)(i) gives

$$\frac{dy}{dx} = \pm \frac{3x^2 + 1}{2\sqrt{x^3 + x}}, \quad x > 0$$

(d)(ii) Hence deduce that the curve $y^2 = x^3 + x$ has no local minimum or maximum points.

(e) The curve has two points of inflexion which, by symmetry, share an x -coordinate. Find the value of that x -coordinate, giving your answer in the form

$$x = \sqrt{\frac{p\sqrt{3} + q}{r}}, \quad p, q, r \in \mathbb{Z}$$

In (d)(ii) an empty list is the answer: there are no such points. In (e) the upper branch $y = \sqrt{x^3 + x}$ carries the inflexion — give the exact value.

```
q1ld = [...]      # every local maximum or
minimum on the curve
q1le = ...         # the x-coordinate of the
points of inflexion

shape = curve(Eq(y**2, x**3 + x))
branch = sqrt(x**3 + x)          #
the upper half, where the question looks

verify_turning('11(d)(ii)', q1ld, shape,
domain=(0.01, 4))
verify_bend('11(e)', q1le, branch, domain=
(0.01, 2))
```

Theory: an argument that runs on parity alone

The hardest questions of this topic never evaluate anything. They establish the **sign** of the gradient on either side of a point and read off the kind.

Suppose $g'_n(x) = nx^{n-1}(b^2 + x^2)$ with $b > 0$ and $n > 1$, and suppose it is already known that $g'_n(1) > 0$ and that $g'_n(0) = 0$. What happens at the origin depends on $g'_n(-1)$:

$$g'_n(-1) = n(-1)^{n-1}(b^2 + 1)$$

The factor $b^2 + 1$ is positive, so the sign is the sign of $(-1)^{n-1}$:

- n **even** $\Rightarrow n - 1$ odd $\Rightarrow (-1)^{n-1} = -1$, so $g'_n(-1) < 0$. The gradient is negative to the left of the origin and positive to the right: the curve falls, flattens, rises — a **local minimum**.
- n **odd** $\Rightarrow n - 1$ even $\Rightarrow (-1)^{n-1} = +1$, so $g'_n(-1) > 0$. The gradient is positive on both sides and zero at the origin: the curve flattens and goes on — an **inflexion with zero gradient**.

Nothing was computed. The second derivative was never written, and for a good reason: at the origin it is zero for every $n > 2$, and the test would have said nothing either way.

One x on each side is enough, provided there is no other stationary point in between — and that is what a given value such as $g'_n(1) > 0$ is for. It fixes the sign on the whole stretch to the right of the origin.

Task 12 — **May 2021 TZ2 Paper 3 Q1(g), 5 marks**

This question explores the family

$$\begin{aligned}f_n(x) &= x^n(a - x)^n, \\a &\in \mathbb{Z}^+, \\n &\in \mathbb{Z}^+, \\n &> 1\end{aligned}$$

Earlier parts give $f'_n(x) = nx^{n-1}(a - 2x)(a - x)^{n-1}$, the three solutions of $f'_n(x) = 0$, and the fact that $f'_n\left(\frac{a}{4}\right) > 0$.

By using that fact and considering the sign of $f'_n(-1)$, show that the point $(0, 0)$ on the graph of $y = f_n(x)$ is

(i) a local minimum point for even values of n , where $n > 1$;

(ii) a point of inflexion with zero gradient for odd values of n , where $n > 1$.

One word each. The check runs the claim over several even and several odd n and over several values of a : a statement about a family has to hold for the family.

```
n, a = symbols('n a', positive=True)

q12_i = ...      # the kind of point at the
origin for even n
q12_ii = ...     # the kind of point at the
origin for odd n

f_n = x**n*(a - x)**n

verify_nature('l2(i)', q12_i, f_n, 0, domain=
(-1, a + 1),
              params={n: (2, 4, 2, 4), a: (2,
2, 3, 5)})
verify_nature('l2(ii)', q12_ii, f_n, 0,
domain=(-1, a + 1),
              params={n: (3, 5, 3, 5), a: (2,
2, 3, 5)})
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
find	find the stationary point, both coordinates
classify	name its kind from the second derivative
sign	name its kind from the sign of the first derivative
inflection	find where the concavity changes
family	the same work with a letter in the function
side	is the point above or below the x -axis
count	for which values of the parameter are there that many
none	show that no such point exists

1. Find the coordinates of the local minimum of $y = x^3 - 3x$.
2. Given $f''(2) = -5$ and $f'(2) = 0$, what kind of point is at $x = 2$?

3. f' changes from negative to positive at $x = 4$. What kind of point is there?
4. Find the x -coordinate of the point of inflexion of $y = xe^{-x}$.
5. Show that $y = x^3 + kx^2$ has a stationary point at $x = -\frac{2k}{3}$.
6. The local minimum of a cubic has $y = -7$. State whether it lies above or below the axis.
7. Find the values of m for which $y = x^3 + mx$ has no stationary point.
8. Show that $y = \frac{1}{x}$ has no local maximum or minimum for $x > 0$.
9. Find every point where the gradient of $y = \sin x + \cos x$ is zero on $[0, 2\pi]$.
0. Where does $y = x^4 - 6x^2$ change from concave down to concave up?
1. $\frac{d^2y}{dx^2} = 6x + 4$ at a stationary point $x = 1$. Maximum or minimum?
2. For which p does $y = x^4 + px^2$ have exactly three stationary points?

```

answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: 'e6640de835ad', 2:
'fc1f263d32b8', 3: '0b6f395ca14a', 4:
'715155bf98ba', 5: 'd34a569ab7aa', 6:
'ea7bf4bb7230', 7: '6c35493a2b93', 8:
'140bedbf9c3f', 9: 'e6640de835ad', 10:
'715155bf98ba', 11: 'fc1f263d32b8', 12:
'6c35493a2b93'})

```

On the clock — *November 2022* *Paper 1 Q7, 7 marks*

Seven marks, ten minutes. No calculator, no hints.

Consider the curve with equation

$$(x^2 + y^2)y^2 = 4x^2, \quad x \geq 0, \quad -2 < y < 2$$

Show that the curve has no local maximum or local minimum points for $x > 0$.

The answer is the list of such points, and it is empty. Write out the implicit

differentiation on paper first — that is where six of the seven marks are.

Attempt log

date	time	result

```
qt = [...]          # every local maximum or
minimum with  $x > 0$ 

shape = curve(Eq(( $x^2 + y^2$ )* $y^2$ ,  $4x^2$ ))

verify_turning('timer', qt, shape, domain=
(0.01, 8))
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 $f'(x) = 4x^2 - 16$, which is zero at $x = \pm 2$; the question wants $p > 0$, so $p = 2$. Then

$$q = f(2) = \frac{4 \cdot 8}{3} - 32 = \frac{32}{3} - 32 = \boxed{-\frac{64}{3}}$$

$f''(x) = 8x$ and $f''(2) = 16 > 0$, so it really is the minimum.

2 (a) The calculator gives the local maximum of $\ln(xe^x + 1) - x^4$ on $[0, 2]$ at $\boxed{(0.709, 0.640)}$ $(0.708519\dots, 0.639580\dots)$.

2 (b) The curve $y = \frac{x(x^2 - 16)}{x^2 + 16}$ has $\boxed{(-1.94, 1.20)}$ as its local maximum and $\boxed{(1.94, -1.20)}$ as its local minimum $(\pm 1.94347\dots, \mp 1.20113\dots)$.

Note the shape: the maximum is the *left* one, because the curve comes up from below on the far left.

3 (b) f' changes from positive to negative at $x = a$, so the curve rises and then falls: a local maximum. (Equivalently $f''(-5) = 6(-5) + 12 = -18 < 0$.)

3 (d) f'' changes sign at $x = c$, so the concavity changes: yes, a point of inflexion. Saying only " $f''(c) = 0$ " earns R0 — that is true at $x = 0$ for $y = x^4$ as well, and there the origin is a minimum.

4 (a) $f'(x) = -2 \sin 2x e^{\cos 2x}$, which is zero when $\sin 2x = 0$, that is $2x = 0, \pi, 2\pi$ and $x = 0, \frac{\pi}{2}, \pi$ inside $-\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$:

$$\left(0, e\right), \quad \left(\frac{\pi}{2}, \frac{1}{e}\right), \quad \left(\pi, e\right)$$

4 (b) $f''(x) = (4 \sin^2 2x - 4 \cos 2x) e^{\cos 2x}$, so

$$\begin{aligned}f''(0) &= -4e < 0, \\f''\left(\frac{\pi}{2}\right) &= \frac{4}{e} > 0, \\f''(\pi) &= -4e < 0\end{aligned}$$

that is maximum, minimum, maximum.

5 With the product and chain rules,

$$f'(x) = \sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}} = \frac{1-2x^2}{\sqrt{1-x^2}}$$

which vanishes at $x = \pm \frac{1}{\sqrt{2}}$, where $f\left(\pm \frac{1}{\sqrt{2}}\right) = \pm \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \pm \frac{1}{2}$. At the ends $f(\pm 1) = 0$, so

$$a = -\frac{1}{2}, \quad b = \frac{1}{2}, \quad \left[-\frac{1}{2} \leq y \leq \frac{1}{2}\right]$$

6 (a) Graphing f' and finding its local minimum (or solving $f'' = 0$) gives $x = -1.60$ ($-1.59537\dots$).

6 (b) Graphing $\frac{d^2y}{dx^2}$ and reading off its zero gives $x = 0.656$ ($0.655996\dots$).

7 (e) $\frac{dy}{dx} = 3x^2 + 2ax = x(3x + 2a)$, zero at $x = 0$ and $x = -\frac{2a}{3}$. At $x = 0$, $y = b$, so $P(0, b)$. At $x = -\frac{2a}{3}$,

$$\begin{aligned}
 y &= \\
 &= -\frac{8a^3}{27} \\
 &+ a \cdot \frac{4a^2}{9} \\
 &+ b \\
 &= -\frac{8a^3}{27} + \frac{12a^3}{27} + b \\
 &= \boxed{\frac{4a^3}{27} + b}
 \end{aligned}$$

7 (f)(i) $\frac{d^2y}{dx^2} = 6x + 2a$. At $x = 0$ it is $2a > 0$, so P is a minimum; at $x = -\frac{2a}{3}$ it is $-4a + 2a = -2a < 0$, so Q is a maximum.

7 (f)(ii) $b > 0$ puts P above the axis, and $\frac{4a^3}{27} + b > 0$ puts Q above it too.

7 (g) Q is below the axis exactly when its y -coordinate is negative:

$$\boxed{\frac{4a^3}{27} + b < 0}$$

8 (i) $V = 4x^3 - 2(a+b)x^2 + abx$, so $\frac{dV}{dx} = 12x^2 - 4(a+b)x + ab$ and

$$\boxed{\frac{d^2V}{dx^2} = 24x - 4(a+b)}$$

8 (ii) Substituting x_m ,

$$\begin{aligned}\left.\frac{d^2V}{dx^2}\right|_{x_m} &= 24 \cdot \frac{(a+b) - \sqrt{a^2 - ab + b^2}}{6} - 4(a+b) \\ &= -4\sqrt{a^2 - ab + b^2} < 0\end{aligned}$$

so x_m gives a local maximum. The $(a+b)$ cancels and what is left is negative for every a and b .

9 (c) $f'(x) = 3x^2 - 3c = 0$ means $x^2 = c$.

- One repeated root, and the gradient touches zero without crossing: $c = 0$.
- Two distinct roots: $c > 0$.
- No real roots: $c < 0$.

9 (d) $x = \pm\sqrt{c}$, and the maximum is the left one:

$$f(-\sqrt{c}) = -c^{3/2} + 3c^{3/2} + 2 = \boxed{2c^{3/2} + 2}$$

$$f(\sqrt{c}) = c^{3/2} - 3c^{3/2} + 2 = \boxed{-2c^{3/2} + 2}$$

10 (e) The minimum value is $-2c^{3/2} + 2$ and the maximum value $2c^{3/2} + 2 > 0$ always, so everything hangs on the minimum.

- One intercept — the minimum is above the axis: $-2c^{3/2} + 2 > 0$, that is $c^{3/2} < 1$ and $\boxed{0 < c < 1}$.
- Two intercepts — the minimum sits on the axis: $\boxed{c = 1}$.
- Three intercepts — the minimum is below the axis: $\boxed{c > 1}$.

10 (f) Two cases.

Case $c \leq 0$. Then $g'(x) = 3x^2 - 3c \geq 0$ for every x , so g never turns and crosses the axis exactly once — for any d at all.

Case $c > 0$. The vertices are $(-\sqrt{c}, 2c^{3/2} + d)$ and $(\sqrt{c}, -2c^{3/2} + d)$, and one intercept means they lie on the same side of the axis:

$$-2c^{3/2} + d > 0 \quad \text{or} \quad 2c^{3/2} + d < 0$$

$$c \leq 0, \quad \text{or} \quad d > 2c^{3/2}, \quad \text{or} \quad d < -2c^{3/2}$$

11 (d)(ii) A quotient is zero only when its numerator is, and $3x^2 + 1 \geq 1 > 0$ for every real x . So $\frac{dy}{dx}$ is never zero and there are no local maxima or minima.

11 (e) Differentiating $2y\frac{dy}{dx} = 3x^2 + 1$ again,

$$2 \left(\frac{dy}{dx} \right)^2 + 2y \frac{d^2y}{dx^2} = 6x$$

At an inflexion $\frac{d^2y}{dx^2} = 0$, so $\left(\frac{dy}{dx} \right)^2 = 3x$, that is

$$\frac{(3x^2 + 1)^2}{4(x^3 + x)} = 3x$$

$$\implies$$

$$12x^2 + 12x^4 = 9x^4 + 6x^2 + 1$$

$$\implies$$

$$3x^4 + 6x^2 - 1 = 0$$

A quadratic in x^2 : $x^2 = \frac{-6 + \sqrt{48}}{6} = \frac{2\sqrt{3} - 3}{3}$, and since $x > 0$,

$$x = \sqrt{\frac{2\sqrt{3} - 3}{3}} \approx 0.393$$

so $p = 2$, $q = -3$, $r = 3$.

12 (i) $f'_n(-1) = n(-1)^{n-1}(a+2)(a+1)^{n-1}$. For even n the exponent $n-1$ is odd, so $(-1)^{n-1} = -1$ and $f'_n(-1) < 0$; with $f'_n(0) = 0$ and $f'_n(\frac{a}{4}) > 0$ the gradient runs $-, 0, +$ across the origin, so $(0, 0)$ is a local minimum.

12 (ii) For odd n the exponent $n-1$ is even, so $(-1)^{n-1} = +1$ and $f'_n(-1) > 0$; the gradient runs $+, 0, +$ and the curve flattens without turning: $(0, 0)$ is a point of inflexion with zero gradient.

The second derivative is useless here — at the origin it vanishes for every $n > 2$, whatever the parity.

Timer

Differentiating $(x^2 + y^2)y^2 = 4x^2$ implicitly, that is $x^2y^2 + y^4 = 4x^2$,

$$2xy^2 + 2x^2y \frac{dy}{dx} + 4y^3 \frac{dy}{dx} = 8x$$

At a local maximum or minimum $\frac{dy}{dx} = 0$, so $2xy^2 = 8x$ and

$$x = 0 \quad \text{or} \quad y^2 = 4$$

The question says $x > 0$ and $-2 < y < 2$, so $x = 0$ is excluded and $y = \pm 2$ is excluded. No candidate survives, and the curve has

no local maximum or minimum points

for $x > 0$.